Lecture 10, Jan 13, 2025.

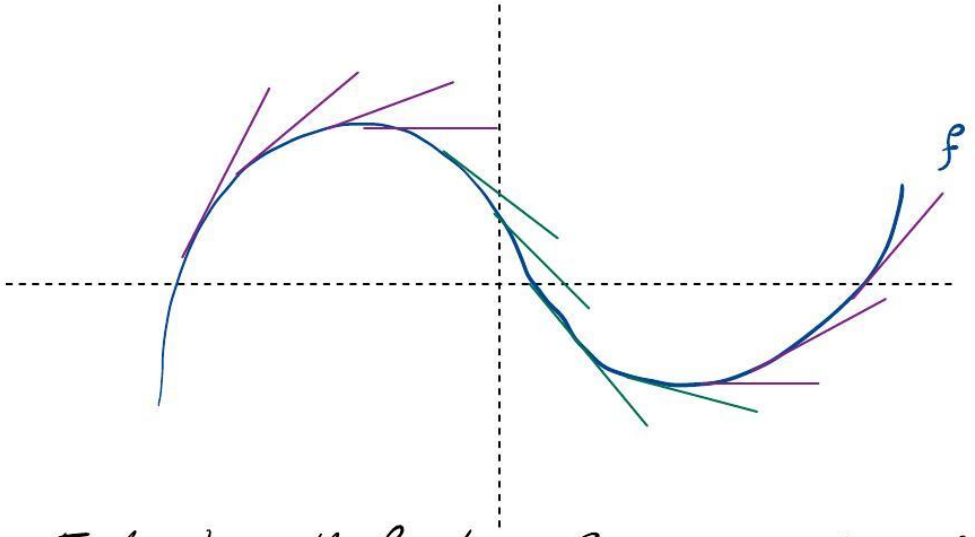
3.3 What Derivatives tell us about the shape of a Graph.

* what Does f' say about f?

Increasing / Decreasing test.

a) If $f'(x)>0$ on an interval, then f is increasing on that interval.

b). If $f'(x)<0$ on an interval, then f is decreasing on that interval.



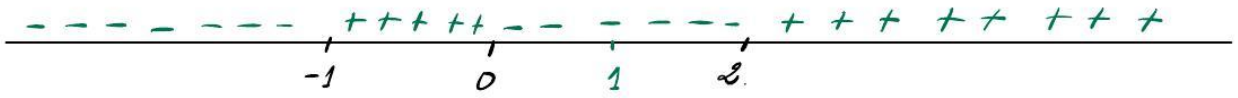
Example. Find where the function $f(x)= 3x^4- 4x^3-12x^2 +5$ is increasing and where is decreasing.

Sln. $f'(x)= 12x^3- 12x^2-24x$

$f'(x)>0 \iff 12x^3- 12x^2-24>0$

$12x(x^2-x-2)>0$

$12x(x-2)(x+1)>0$



$f(x)$ is increasing on $(-1,0)\cup(2,\infty)$.

$f(x)$ is decreasing on $(-\infty,-1)\cup(0,2)$.

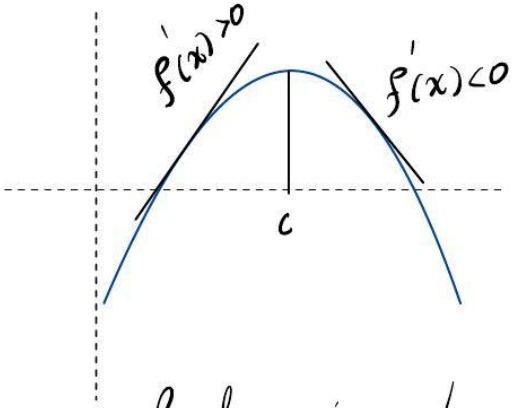
The first Derivative test

The first Derivative test: Suppose that c is a critical number of a continuous function f.

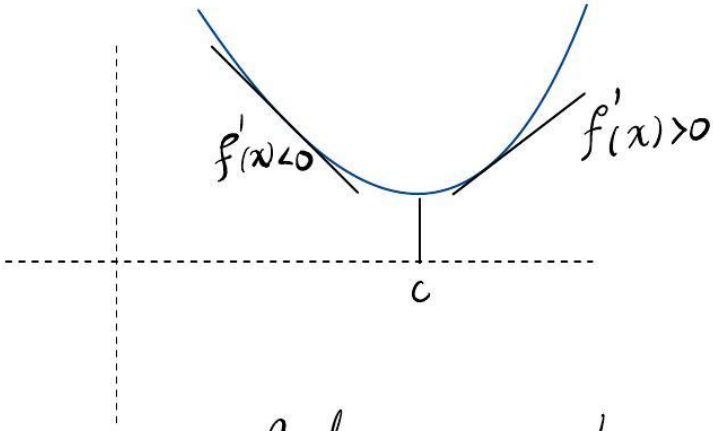
a) If f' changes from (+) to (-) at c, then f has a local maximum

b). If f' changes from (-) to (+) at c, then f has a local minimum.

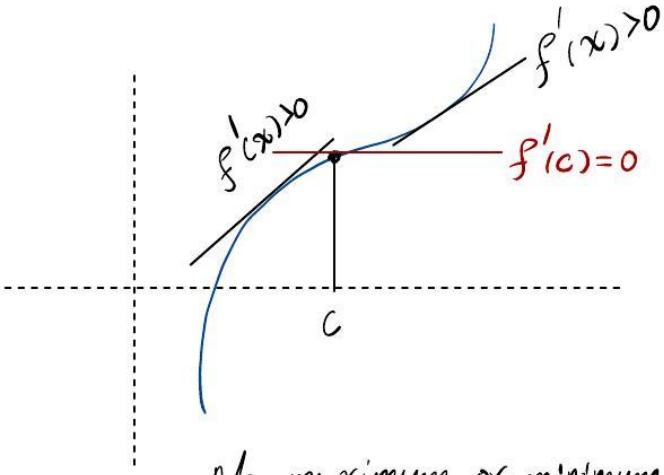
c). If f' is positive to the left and right of c, or negative to the left and right of c, then f has no local minimum or maximum.



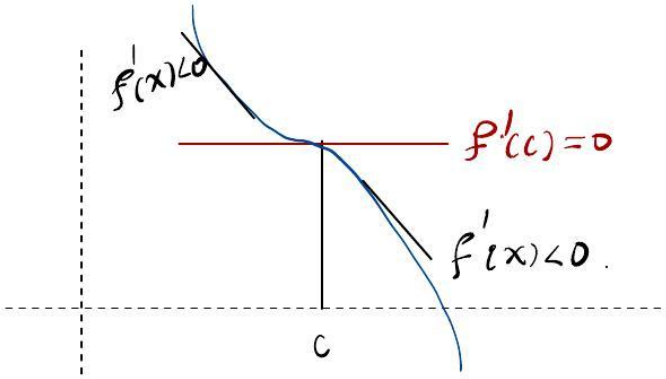
local maximum at c.



local minimum at c.



No maximum or minimum



No maximum or minimum

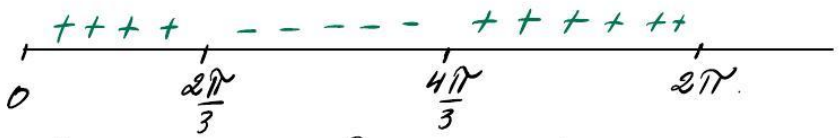
Example. Find the local maximum and minimum values of the function

$$g(x) = x + 2\sin x, \quad 0 \le x \le 2\pi.$$

Sln. We need the critical values: then, the derivative is.

$$g'(x) = 1 + 2\cos x.$$

So, $g'(x) = 0 \iff \cos x = -\frac{1}{2}$. the solution of this equation are $\frac{2\pi}{3}$, $\frac{4\pi}{3}$ then, computing the maximum and minimum.



Because $g'(x)$ change from (+) to (-) at $\frac{2\pi}{3}$, the First Derivative test tell us there is a local maximum at $\frac{2\pi}{3}$ and the local maximum is

$$g\left(\frac{2\pi}{3}\right) = \frac{2\pi}{3} + 2\sin\left(\frac{2\pi}{3}\right) = \frac{2\pi}{3} + 2\left(\frac{\sqrt{3}}{2}\right) = \frac{2\pi}{3} + \sqrt{3} \approx 3.83.$$

Likewise, $g'(x)$ changes from (-) to (+) at $\frac{4\pi}{3}$, then

$$g\left(\frac{4\pi}{3}\right) = \frac{4\pi}{3} + 2\sin\left(\frac{4\pi}{3}\right) = \frac{4\pi}{3} - \sqrt{3} \approx 2.46.$$

is a local minimum.

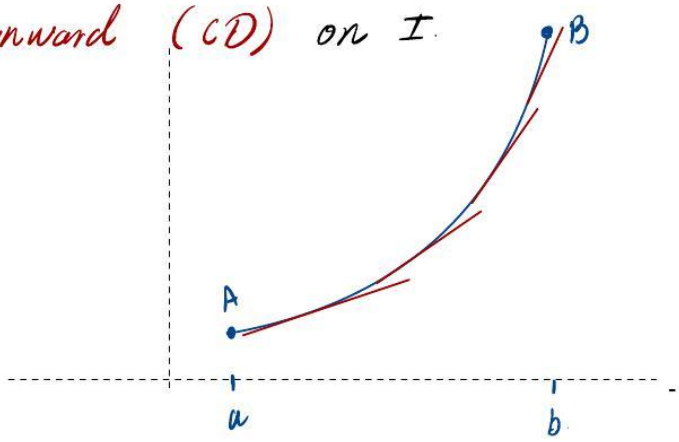
$g(0) = 0.$

$g(2\pi) = 2\pi + 2\sin(2\pi) = 2\pi.$

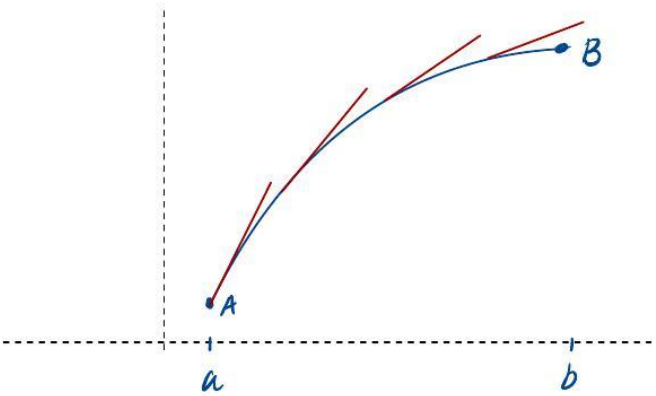
So, the absolute maximum is 2π and absolute minimum is 0.

* What Does f'' say about f?

Def. If the graph of f lies above all of its tangents on an interval I, then f is called concave upward (CU) on I and if the graph of f lies below of its tangents on I, then f is called concave downward (CD) on I.



a). Concave upward (CU)



b). Concave downward (CD).

Concavity test

a). If $f''(x)>0$ on an interval I, then the graph of f is (CU) on I.

b). If $f''(x)<0$ on an interval I, then the graph of f is (CD) on I.

Def. A point P on a curve $y=f(x)$ is called an inflection point if f is continuous there are a curve changes from (CU) to (CD) or from (CD) to (CU)

Rmk. $f''(x)$ changes from (+) to (-) or (-) to (+).

Example Sketch a possible graph of a function f that satisfies the following conditions.

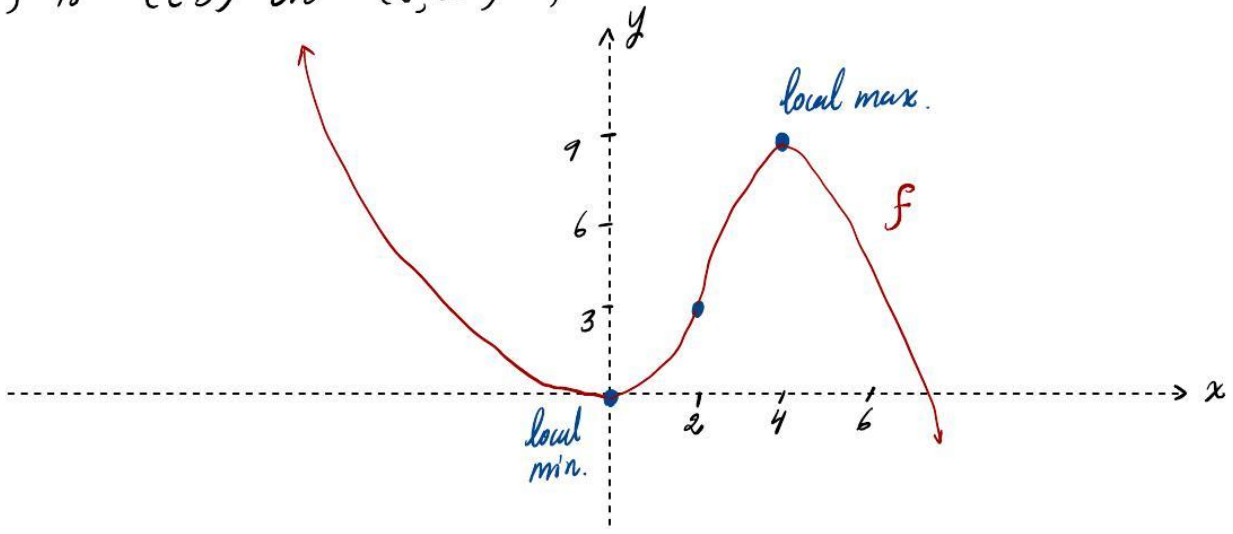
a). $f(0)=0$, $f(2)=3$, $f(4)=6$, $f'(0)=f'(4)=0$.

b). $f'(x)>0$ for $0<x<4$ $f'(x)$ for $x<0$ and $x>4$.

c). $f''(x)>0$ for $x<2$, $f''(x)<0$ for $x>2$.

Sln. f is increasing on $(0,4)$
f is decreasing on $(-\infty,0)\cup(4,\infty)$ } $x=0$ we have a local min.
$x=4$ we have a local max.

f is (CU) on $(-\infty,2)$
f is (CD) on $(2,\infty)$ } $x=2$ is an inflection point.



The Second Derivative test. Suppose f'' is continuous near c.

a). If $f'(c)=0$ and $f''(c)>0$ then f has a local minimum at c.

b). If $f'(c)=0$ and $f''(c)<0$ then f has a local maximum at c.

Rmk. The second derivative test is inconclusive when $f''(c)=0$. In other words, at such a point there might be a maximum, the might a minimum or there might be neither.

Example (for you). Discuss the curve $y=x^4-4x^3$ with respect to concavity points of inflection, and local min/max.

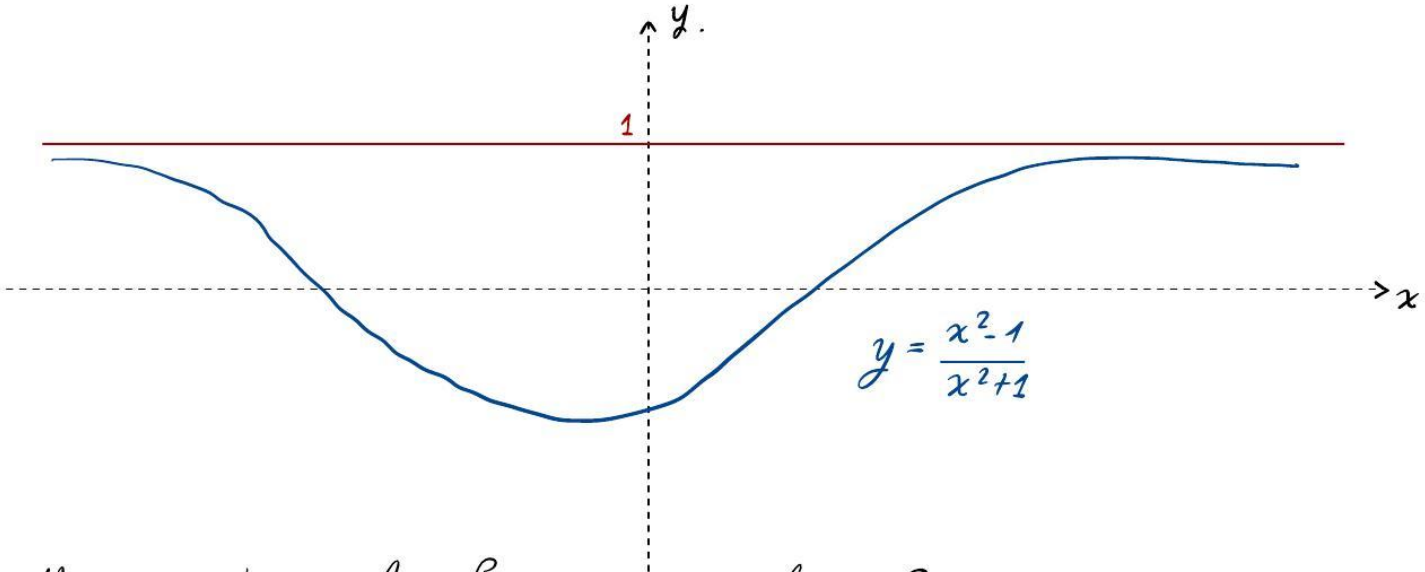
3.4. Limits at infinity; Horizontal Asymtotes.

Let consider

$$f(x)=\frac{x^2-1}{x^2+1}$$

Using the next Data.

x	0	± 1	± 3	± 10	± 100	- - - - $x\to\infty$
$f(x)$	-1	0	0.8	0.98019	0.999800	- - - - $f(x)\to 1$



then, we have $\lim_{x\to\infty} f(x)=1$ and $\lim_{x\to-\infty} f(x)=1$.

Def. Let f be a function defined on an interval (a,∞). Then,

$$\lim_{x\to\infty} f(x) = L.$$

means that the values of $f(x)$ can be made arbitrarily close to L by taking x to be sufficiently large.

Def. the line $y = L$ is called a horizontal asymtote of the curve $y = f(x)$ if either $\lim_{x\to\infty} f(x) = L$ or $\lim_{x\to-\infty} f(x) = L$.

Thm. If $r>0$ is rational number then

$$\lim_{x\to\infty} \frac{1}{x^r} = 0.$$

If $r>0$ is a rational number s.t x^r is defined for all x, then

$$\lim_{x\to-\infty} \frac{1}{x^r} = 0.$$

Example. Evaluate the following limit.

$$\lim_{x\to\infty} \frac{3x^2 - x - 2}{5x^2 + 4x + 1}.$$

Sln.

$$\lim_{x\to\infty} \frac{3x^2 - x - 2}{5x^2 + 4x + 1} = \lim_{x\to\infty} \frac{\frac{3x^2 - x - 2}{x^2}}{\frac{5x^2 + 4x + 1}{x^2}}$$

$$= \lim_{x\to\infty} \frac{\frac{3x^2}{x^2} - \frac{x}{x^2}\frac{x^2}{} - \frac{2}{x^2}}{\frac{5x^2}{x^2} + \frac{4x}{x^2} + \frac{1}{x^2}}$$

$$= \lim_{x\to\infty} \frac{3 - \frac{1}{x} - \frac{2}{x^2}}{5 + \frac{4}{x} + \frac{1}{x^2}}$$ By the thm.

$$= \frac{3}{5}.$$

then, $y = \frac{3}{5}$ is an horizontal asymtote.

If degree Numerator = degree Denominator $\Rightarrow$ the limit is a number.

If degree Numerator > degree Denominator $\Rightarrow$ the limit DNE ∞ or $-\infty$.

If degree Numerator < degree Denominator $\Rightarrow$ the limit exist and is equal to 0.

Example. Find the horizontal asymptotes of the graph of the function

$$f(x) = \frac{\sqrt{2x^2+1}}{3x-5}$$

Sln. • Identify the large power in the Denominator or Numerator.

• Divides the numerator and denominator by this power.

The large power is x.

$$\lim_{x\to\infty} \frac{\sqrt{2x^2+1}}{3x-5} = \lim_{x\to\infty} \frac{\frac{\sqrt{2x^2+1}}{x}}{\frac{3x-5}{x}}$$

$$= \lim_{x\to\infty} \frac{\sqrt{\frac{2x^2+1}{x^2}}}{\frac{3x}{x} - \frac{5}{x}}$$

$$= \lim_{x\to\infty} \frac{\sqrt{2 + 1/x^2}}{3 - 5/x} = \frac{\sqrt{2}}{3}.$$

So, $y = \frac{\sqrt{2}}{3}$ is a horizontal Asymptote of the graph of f.

$$\lim_{x\to-\infty} \frac{\sqrt{2x^2+1}}{3x-5} = \lim_{x\to-\infty} \frac{\frac{\sqrt{2x^2+1}}{-x}}{\frac{3x-5}{x}}$$

b/c $|x| = -x$ $\sqrt{x^2} = |x|$

$$= \lim_{x\to-\infty} \frac{-\sqrt{\frac{2x^2+1}{x^2}}}{\frac{3x}{x} - 5/x} = \lim_{x\to-\infty} \frac{-\sqrt{2 + 1/x^2}}{3 - 5/x} = -\frac{\sqrt{2}}{3}$$

So, $y = -\frac{\sqrt{2}}{3}$ is a horizontal asymtote.

Example. Compute $\lim_{x\to\infty} \left(\sqrt{x^2+1} - x\right)$.

Sln.

$$\lim_{x\to\infty} \left(\sqrt{x^2+1} - x\right) \times \frac{\left(\sqrt{x^2+1} + x\right)}{\left(\sqrt{x^2+1} + x\right)} = \lim_{x\to\infty} \frac{x^2+1-x^2}{\sqrt{x^2+1}+x} = \lim_{x\to\infty} \frac{1}{\sqrt{x^2+1}+x} = 0$$

Example. Compute $\lim_{x\to\infty} \sin\left(\frac{1}{x}\right)$.

Sln. Since $\frac{1}{x} \to 0$ as $x \to \infty$ then $\sin\left(\frac{1}{x}\right) \longrightarrow \sin(0)$ as $x \to \infty$.

$$\lim_{x\to\infty} \sin\left(\frac{1}{x}\right) = \sin(0) = 0.$$

Infinite Limits at Infinity

The notation $\lim_{x\to\infty} f(x) = \infty$ is used to indicate that the values of $f(x)$ becomes large as x becomes large. Similarly for

$$\lim_{x\to-\infty} f(x) = \infty, \quad \lim_{x\to\infty} f(x) = -\infty, \quad \lim_{x\to-\infty} f(x) = -\infty.$$

Example. Find $\lim_{x\to\infty} x^3$ and $\lim_{x\to-\infty} x^3$.

Sln. $\lim_{x\to\infty} x^3 = \infty \qquad \lim_{x\to-\infty} x^3 = -\infty.$

Example. $\lim_{x\to\infty} (x^2 - x)$.

Sln. $\lim_{x\to\infty} (x^2-x) = \lim_{x\to\infty} x(x-1) = \infty.$

Example. $\lim_{x\to\infty} \frac{x^2+x}{3-x} = \lim_{x\to\infty} \frac{x(x+1)}{x\left(\frac{3}{x}-1\right)} = \lim_{x\to\infty} \frac{x+1}{\frac{3}{x}-1} = -\infty.$

Example. Sketch the graph of $y = (x-2)^4(x+1)^3(x-1)$ by finding its intercepts and its limits as $x\to\infty$ and as $x\to-\infty$.

Sln. For y-intercepts: $f(0) = (-2)^4(1)^3(-1) = -16$; $(0, -16)$.

x-intercepts: $0 = (x-2)^4(x+1)(x-1) \Rightarrow x = 2, x = \pm 1$.

So, $(2,0)$, $(1,0)$, $(-1,0)$ are the x-intercepts.

$\lim_{x \to \infty} (x-2)^4(x+1)(x-1) = \infty$

$\lim_{x \to -\infty} (x-2)^4(x+1)(x-1) = \infty$.

